

Drugs In Pregnancy

Prepared by:

Dalia Ghareb

Maulud

Liya saeed Ahmad

Sara Hama Abdulla

Chya Nasir qadir



Introduction

Pregnancy is a unique and complex period in a woman's life. During this time, many physiological changes occur in the body that can significantly affect how drugs are absorbed, distributed, metabolized, and eliminated.



Physiological Changes Affecting Drug Pharmacology

- Gastric PH & Gastric Motility
- Maternal Plasma Volume
- Hormonal changes
- Glomerular Filtration Rate

Placental Transfer and Fetal Exposure

The placenta functions as an organ of exchange, allowing nutrients — and unfortunately, some medications — to pass from the mother to the fetus.

Drugs Cross placenta :

- low in molecular weight
- poorly protein-bound
- highly lipophilic
- low in ionization



The first trimester of pregnancy (0–13 weeks) is the most critical period for embryonic and fetal development, and the time when the risk of teratogenic effects is greatest.

Medication Use and Compliance in Pregnancy

- 40–90% of pregnant women are exposed to medications during pregnancy.
- drugs should be avoided unless absolutely necessary (except serious or life threatening conditions.
- compliance among pregnant women tends to be suboptimal(39% noncompliance) .

How Drugs Affect the Fetus

Drugs can affect the fetus in three main ways:

1. Direct Effects — Causing injury, abnormal development (leading to birth defects), or even fetal death.

2. Placental Effects — Some drugs constrict placental blood vessels, reducing oxygen and nutrient delivery, which can lead to an underweight or underdeveloped baby.

3. Uterine Effects — Certain drugs can stimulate uterine contractions, reducing fetal blood supply or triggering preterm labor and delivery.



Drug Safety During Pregnancy and Breastfeeding

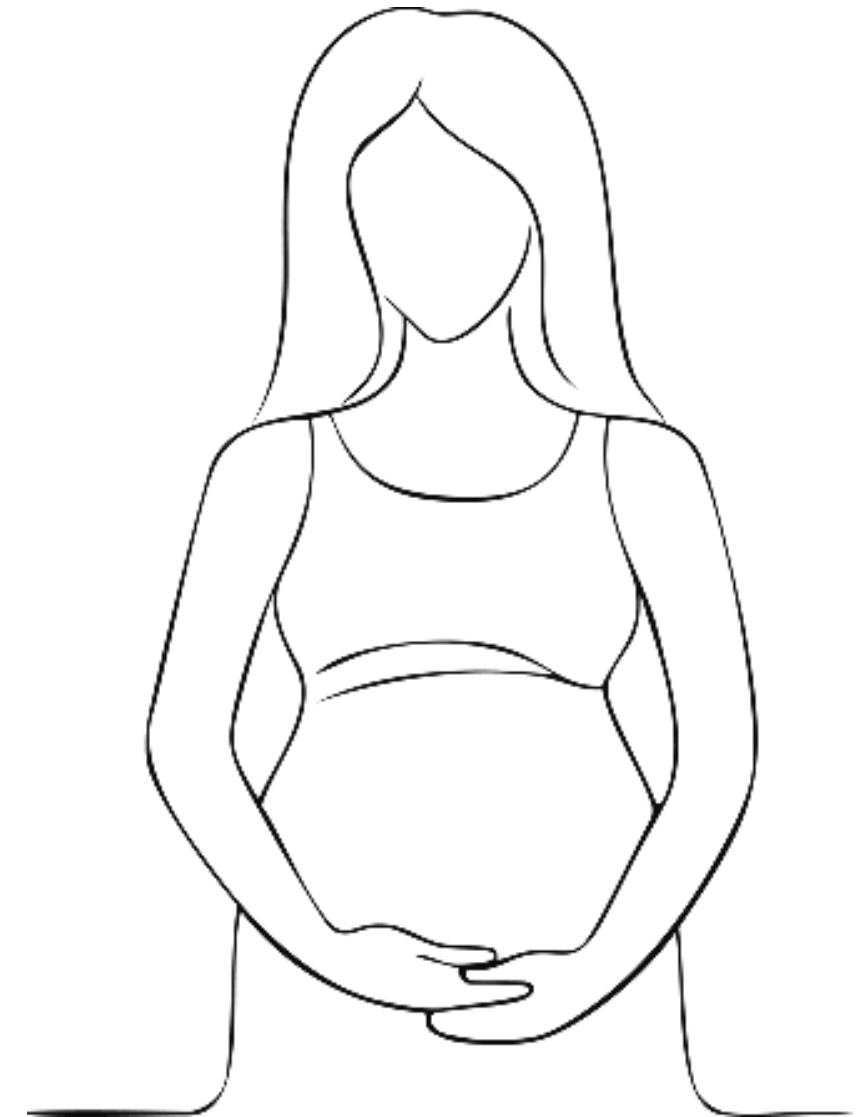
- After birth, medications can also pass into breast milk in small amounts.

Most are safe, but some — such as cytotoxic agents, lithium, radiopharmaceuticals, and retinoids — must be strictly avoided while breastfeeding.

- The U.S. Food and Drug Administration (FDA) , Pregnancy and Lactation Labeling Rule (PLLR)
- benefit–risk assessment must be carefully evaluated to ensure maternal therapeutic benefit while minimizing fetal risk.

Teratogenicity & Drug Safety Classification

- **Teratogenicity** means the ability of a drug to cause structural or functional abnormalities in the developing embryo or fetus.
- Teratogenic effects can lead to miscarriage, stillbirth, growth restriction, or birth defects such as limb deformities or neural-tube defects.
- Drug causing this effect called **Teratogen** , examples :
 - **Thalidomide** → limb deformities
 - **Valproic acid** → neural-tube defects
 - **ACE inhibitors** → fetal renal damage



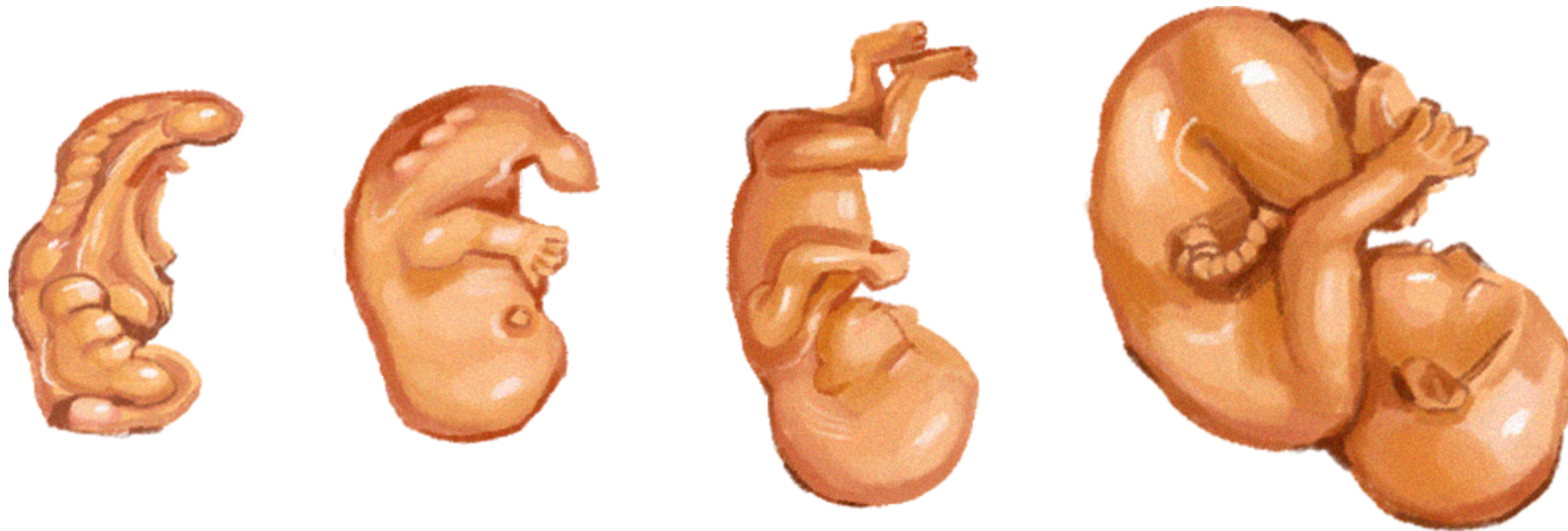


Malformations due to maternal ingestion of thalidomide (Schardein 1982 and Moore 1993).

thalidomide tragedy in the 1950s–60s, where a morning-sickness drug caused thousands of malformed births, completely changed drug-safety regulations.

Critical Periods of Fetal Development

- The most critical period of fetal development is the **first trimester** during which major organs and the basic body structure are formed.
- During this time, the developing fetus is most vulnerable to teratogens that can cause structural birth defects.



Critical Periods of Fetal Development

Stage	Time(weeks)	Effect if exposed
Pre-implantation	0-2 weeks	“All or non “ efect (either death or survival)
Organogenesis	3-8 weeks	Major malformations (heart, brain,limbs)
Fetal period	9 weeks - term	Growth retardation , functional deficits

FDA Pregnancy Risk Categories (A–X)

Table 1. **Current FDA Pregnancy Categories**

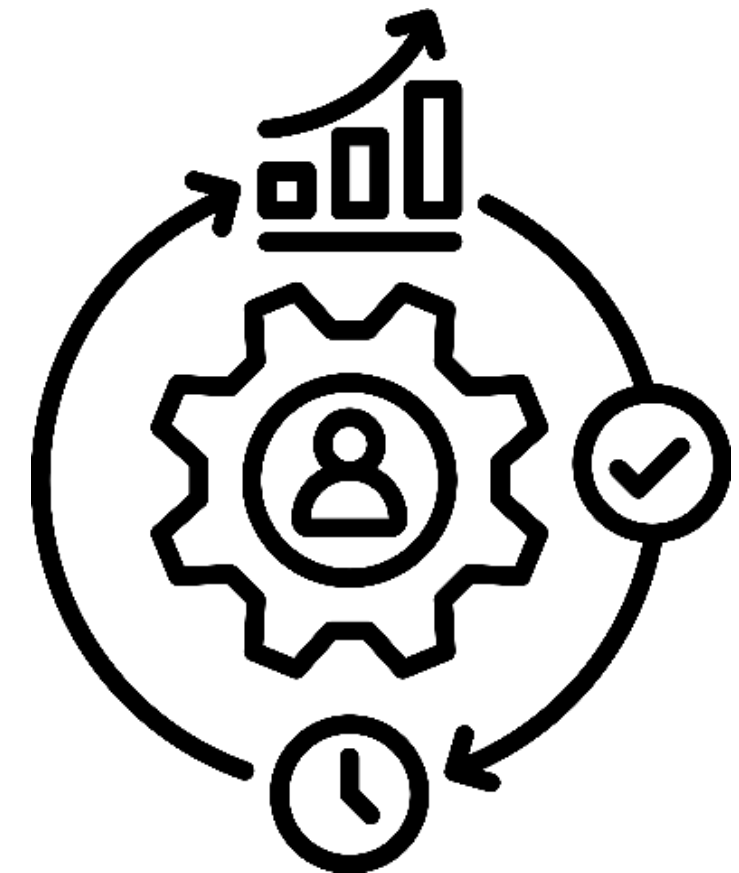
Category	Description
A	Well-controlled studies in humans show no risk to the fetus
B	No well-controlled studies have been conducted in humans; animal studies show no risk to the fetus
C	No well-controlled studies have been conducted in humans; animal studies have demonstrated an adverse effect on the fetus
D	Evidence of human risk to the fetus exists; however, benefits may outweigh risks in certain situations
X	Controlled studies in animals or humans demonstrate fetal abnormalities; the risk in pregnant women clearly outweighs any possible benefit

Source: Reference 1.

But this letter system was too simple, and in 2015 it was officially replaced.

New System: PLLR (Pregnancy and Lactation Labeling Rule)

- since June 2015, the FDA replaced the old category system with the **Pregnancy and Lactation Labeling Rule (PLLR)** because the letter categories were overly simple and often misleading.
- Instead of a single letter, it provides clear, evidence-based details in three sections:



New System: PLLR (Pregnancy and Lactation Labeling Rule)

1. **Pregnancy**: risk summary, clinical considerations, labor & delivery info.
 2. **Lactation**: drug levels in milk and potential infant effects.
 3. **Females & Males of Reproductive Potential**: fertility, contraception, pregnancy testing.
- This system helps healthcare professionals make individualized, informed decisions rather than relying on a single letter."

Common Conditions & Safe Drug Choices

Common conditions

- Hypertension
- Diabetes
- Nausea & Vomiting
- Epilepsy

Hypertension

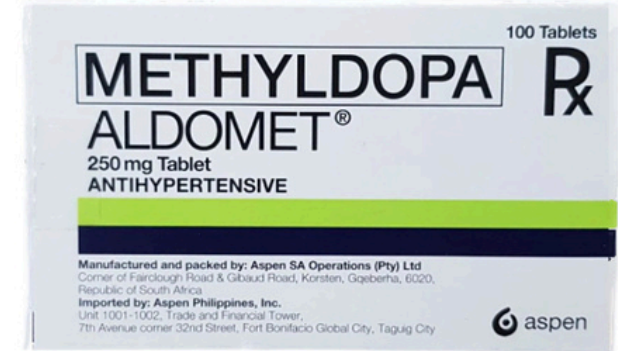
- Hypertension is common in pregnancy. Approximately 1 in 10 pregnancies will be complicated by one or more episodes

Classification

1. Gestational
2. Pre-eclampsia
3. Chronic hypertension

Medications for hypertension

- **Labetalol**
 - Combined alpha and beta blocking agent
 - starting dose of 100 mg 2–3 times daily
- **Methyldopa**
 - Centrally acting antihypertensive agent
 - Available in oral tablet of 250mg
- **Nifedipine**
 - Calcium-channel blocker with a rapid onset of action.
 - 10-20mg orally ; maximum 120mg per day



Hydralazine (Emergency Use)

- Used for severe hypertension in pregnancy.
- Initial dose: 5 mg IV given slowly.
- If BP still high: Repeat 5 mg every 20–30 minutes, up to 15 mg total.
- If still uncontrolled: Start IV infusion (e.g., 20 mg in 200 mL normal saline over 30 min).

Diabetes

Mellitus

Diabetes may complicate a pregnancy either because a woman has:

- Pre-existing diabetes (Type 1 or Type 2)
- Gestational Diabetes Mellitus (GDM)



Management of Diabetes Mellitus

- Insulin – preferred and safest option for controlling blood glucose during pregnancy.
- 1st trimester: ~0.6 units/kg/day
- 2nd trimester: ~0.7 units/kg/day
- 3rd trimester: ~0.8-1.0 units/kg/day



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Medications

- Metformin – may be used if blood glucose isn't controlled with diet and exercise or add to insulin If needed
- Start with: 500 mg once daily with or after food then increase it gradually as tolerated.



Nausea and Vomiting

- Very common (70-80%)

Treatment

- Start with lifestyle & non-drug measures
- First-line: Doxylamine + Pyridoxine
- Second-line: Metoclopramide (<5 days)

Nausea and Vomiting

- Nausea and vomiting in pregnancy are extremely common; 70–80% of pregnant women experience these symptoms
- Often start with non-drug/lifestyle measures. If drug needed, the first-line would be Doxylamine/pyridoxine.
- If mild measures + first line drug insufficient, escalate to second-line options such as Metoclopramide (limited to short-term use (≤ 5 days) due to side-effects.)

Hyperemesis gravidarum

Supportive Care

- IV fluids (Normal saline + potassium if needed)
- Thiamine supplementation before giving dextrose
- Correct electrolyte imbalance

Antiemetics

- First line: Cyclizine / Promethazine
- Second line: Metoclopramide / Ondansetron

Epilepsy

- Epilepsy is the most common serious neurological condition in pregnancy
- All AEDs carry some fetal risk, but careful selection and monitoring reduce this significantly.

Management of Epilepsy

- **Lamotrigine:** generally considered first-line in pregnancy
- **Carbamazepine:** Acceptable with caution Slightly increased risk of neural tube defects; folic acid supplementation recommended (5 mg/day)
- **Levetiracetam:** safe and common, low teratogenic risk

Contraindicated drugs during pregnancy

Organogenesis: first trimester

drugs that cause congenital malformations

Isotretinoin (Retinoic acid)

- Cause: Disrupts neural crest cell migration; teratogenic retinoic-acid receptor activation.
 - Effect: Craniofacial defects, cardiac malformations, CNS anomalies, thymic aplasia → retinoid embryopathy.

Methotrexate

- Cause: Folate antagonist → inhibits DNA synthesis and cell division.
- Effect: Skeletal anomalies, limb defects, neural tube defects, growth restriction.

Thalidomide

- Cause: Anti-angiogenic effects impair limb bud development.
- Effect: Limb reduction defects (phocomelia), ear deformities, heart defects.

ACE inhibitors / ARBs

- Cause: Interferes with fetal renin-angiotensin system.
- Effect: Cardiac and CNS malformations when used in first trimester.

Valproic Acid

- Cause: Histone deacetylase inhibition → impaired neural tube closure.
- Effect: Spina bifida, cardiac defects, craniofacial anomalies.

Warfarin

- Cause: Vitamin K antagonism → impaired cartilage and bone formation.
- Effect: Nasal hypoplasia, stippled epiphyses (“fetal warfarin syndrome”).

Second Trimester

Drugs that impair fetal development or function rather than cause structural defects

ACE inhibitors / ARBs

- Cause: Fetal renal tubular dysplasia from renin-angiotensin suppression.
- Effect: Oligohydramnios, fetal anuria, renal failure.

NSAIDs (after 20 weeks)

- Cause: Prostaglandin inhibition.
- Effect: Fetal kidney dysfunction → oligohydramnios.

Tetracyclines

- Cause: Bind fetal bone and teeth.
- Effect: Permanent yellow-brown tooth discoloration, growth delay.

Fluoroquinolones

Third Trimester

Drugs that affect near-term physiology

NSAIDs (especially after 30 weeks)

- Cause: Inhibition of prostaglandin-dependent ductal patency.
- Effect: Premature closure of ductus arteriosus, pulmonary hypertension.

Opioids (chronic use)

- Cause: Placental transfer causing sustained fetal exposure.
- Effect: Neonatal opioid withdrawal syndrome (NOWS).

SSRIs – especially paroxetine

- Cause: Serotonergic transfer; late-term exposure.
- Effect: Persistent pulmonary hypertension of the newborn (PPHN), neonatal adaptation syndrome.

Benzodiazepines (high-dose or chronic)

- Cause: CNS depressant placental transfer.
- Effect: Neonatal sedation, withdrawal, hypotonia (“floppy infant syndrome”).

Contraindicated Drugs During Breastfeeding

1. Cytotoxic / Immunosuppressive Agents

Methotrexate

- **Cause:** Accumulates in breast milk; inhibits folate metabolism in rapidly dividing infant cells.
- **Effect:** Bone marrow suppression, immune suppression, potential growth impairment.

Cyclophosphamide

- **Cause:** Alkylating agent → mutagenic and cytotoxic in infants; excreted in milk.
- **Effect:** Neutropenia, immunosuppression, possible carcinogenic risk.

Chlorambucil

- **Cause:** DNA alkylation in infant cells.
- **Effect:** Risk of bone marrow suppression.

2. Radioactive Substances

Radioactive Iodine (I-131)

- Cause: Concentrates in breast milk + infant thyroid tissue.
- Effect: Infant thyroid ablation → hypothyroidism, potential thyroid cancer risk.

Technetium-99m (for certain radiopharmaceuticals)

- **Cause:** Milk contamination with radionuclides.
- **Effect:** Radiation exposure in infant → avoid breastfeeding for the recommended clearance period.

3. Psychotropic & CNS-Active Drugs

Lithium

- Cause: High milk-to-plasma ratio; infant excretion is poor → accumulation.
- Effect: Hypotonia, cyanosis, arrhythmias, nephrotoxicity, thyroid dysfunction.

Clozapine

- **Cause:** Lipophilic antipsychotic → high levels in milk; infant CYP metabolism immature.
- **Effect:** Agranulocytosis risk, sedation, seizures, apnea.

High-dose benzodiazepines (chronic use)

- **Cause:** Neonatal accumulation due to immature hepatic clearance.
- **Effect:** Lethargy, poor feeding, respiratory depression.

4. Drugs of abuse

Cocaine

- **Cause:** High concentration in milk; potent CNS stimulant.
- **Effect:** Irritability, seizures, tachycardia, hypertension, neurotoxicity.

Methamphetamine / Amphetamines (abuse)

- **Cause:** Concentrate in milk; slow neonatal elimination.
- **Effect:** Poor weight gain, irritability, tachycardia, neurodevelopmental impact.

Alcohol (heavy use)

- **Cause:** Direct transfer into milk proportional to maternal serum levels.
- **Effect:** Sleep disturbances, impaired motor development, decreased milk production

6. Cardiovascular & Endocrine Drugs

Amiodarone

- **Cause:** Contains iodine + high fat solubility → concentrates in milk.
- **Effect:** Infant hypothyroidism, bradycardia.

Ergot Alkaloids (ergotamine, methylergonovine in high doses)

- **Cause:** Vasoconstriction and suppression of prolactin.
- **Effect:** Infant vomiting, hypertension, convulsions; decreased milk supply.

7. Hormonal Agents

Estrogen-containing contraceptives (early postpartum)

- **Cause:** Suppress prolactin and milk production.
- **Effect:** Reduced milk volume; no major infant toxicity but **contraindicated until 3–6 weeks postpartum**

8. Miscellaneous

Retinoids (oral isotretinoin)

- Cause: Unknown if safe; potential high lipophilicity and toxicity.
- Effect: Breastfeeding contraindicated by FDA due to theoretical teratogenic risk.

Aspirin (chronic high doses)

- Cause: Salicylate accumulation.
- Effect: Metabolic acidosis, bleeding tendency, possible Reye syndrome

Conclusion

Drug use in pregnancy must be carefully limited because many medications cross the placenta and can harm the fetus—especially in the first trimester, when organ development occurs. Choose safe alternatives when possible, avoid known teratogens, and use the PLLR system for risk assessment.

Always balance maternal benefit vs fetal risk, and be cautious during breastfeeding since some drugs enter breast milk.

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**Thank
you**